INCH-POUND

MIL-DTL-17361/3A(SH)
19 September 2006

SUPERSEDING
MIL-C-17361/3(SH)
10 March 1989

DETAIL SPECIFICATION SHEET

CIRCUIT BREAKERS, AIR, ELECTRIC, INSULATED HOUSING (SHIPBOARD USE) TYPE AQB-LF100 (FOR REPLACEMENT ONLY)

This document remains inactive for new design.

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17361.

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TABLE I. Requirements.

Maximum rated continuous current	100 amperes
Voltage and frequency	500 VAC, 60 Hz 250 VDC
Interrupting rating	100,000 amperes, 60 Hz, AC 150,000 amperes, 250 VDC 150,000 amperes, 355 VDC
Peak let-through current	Not applicable
Trip unit	Interchangeable
Current limiting fuses	LP5H100RE, 5,000–15,000 amperes LP5H100RD, 10,000–15,000 amperes
Type of connections	Back (removable mtg.)
Electric operator	Not available
Undervoltage trip or shunt trip	Neither available
Auxiliary switch	Not available
Handle lock	Group 2 (Figure 1 of MIL-DTL-17361)
Handle boot	Not available

NOTE: Type AQB-LF100 circuit breakers shall be used for replacement only.

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TABLE II. Instantaneous trip calibrations (all 3 poles).

Fuse style	Trip element rating (amps) /1	Instantaneous trip settings – amperes/maximum time	
		Current (amperes) /2	Time (seconds)
LF5H100RE	15D	195	0.024
	25D	325	
	25DF	195	
LF5H100RD	50D	650	0.024
	50DF	325	
	75D	975	
	75DF	487	
	100D	1300	
	100DF	650	
	100DG	2000	

NOTES:

/1 Identification marking of trip units:
Numerals indicate continuous current rating of the specific trip unit.
Letter “D” denotes that the trip units are applicable to circuit breaker type AQB-LF100.
Letter “F” denotes calibration for applications having low inrush currents.
Letter “G” denotes special calibration for use as generator or transformer protection.

/2 The tolerance for instantaneous trip shall not exceed ±10%. Time shall be measured at three times pickup current. For qualification tests only, not for routine calibration.

TABLE III. Tripping characteristics – long-time delay (LTD) (all 3 poles).

Current as multiple of trip element rating	Tripping time 60-Hz AC trip element
1.30	Not less than 1 hour
2.25	Less than 1 hour
6.00	Between 20 and 32 seconds

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TABLE IV. Interruption performance test requirements /1.

Circuit sample	Element amperes	Interrupting amperes	Power factor lagging	Unit operation	See note
1A	100	100,000	0.15–0.20	O-CO-CO	/2
1A (1)	100	10,000	0.45–0.50	O-CO-CO	/3 /4
1A (1)		15,000	0.25–0.30	O-CO-CO	/2 /3
1A (1)		25,000	0.15–0.20	O-CO-CO	/2 /3
1A (1)		50,000	0.15–0.20	O-CO-CO	/2 /3
1B	15	100,000	0.15–0.20	O-CO-CO	/2 /3
1B (1)	25	5,000	0.45–0.50	O-CO-CO	/3 /4
1B (1)		15,000	0.25–0.30	O-CO-CO	2/ 3/
1B (1)		25,000	0.15–0.20	O-CO-CO	/2 /3
1B (1)		50,000	0.15–0.20	O-CO-CO	/2 /3
1B (2)	50	100,000	0.15–0.20	O-CO-CO	/2 /3

NOTES:

/1 Oscillograms shall be taken on all interrupting tests.

/2 Circuit breaker shall open and at least one fuse blow.

/3 New circuit breakers may be used for test above the standard circuit breaker rating.

/4 Circuit breaker shall open on instantaneous trip and fuses shall not blow.

TABLE V. Endurance and temperature rise test requirements.

Electrical endurance test	The rate of operation shall be 6 operations per minute for a total of 6000 close-open operations.		
Temperature rise tests	Size of cable and bus bars for temperature rise tests		
	Copper rating (amperes)	Cable size	Copper bus size (inch)
	100	AWG 3	3/4 X 1/8

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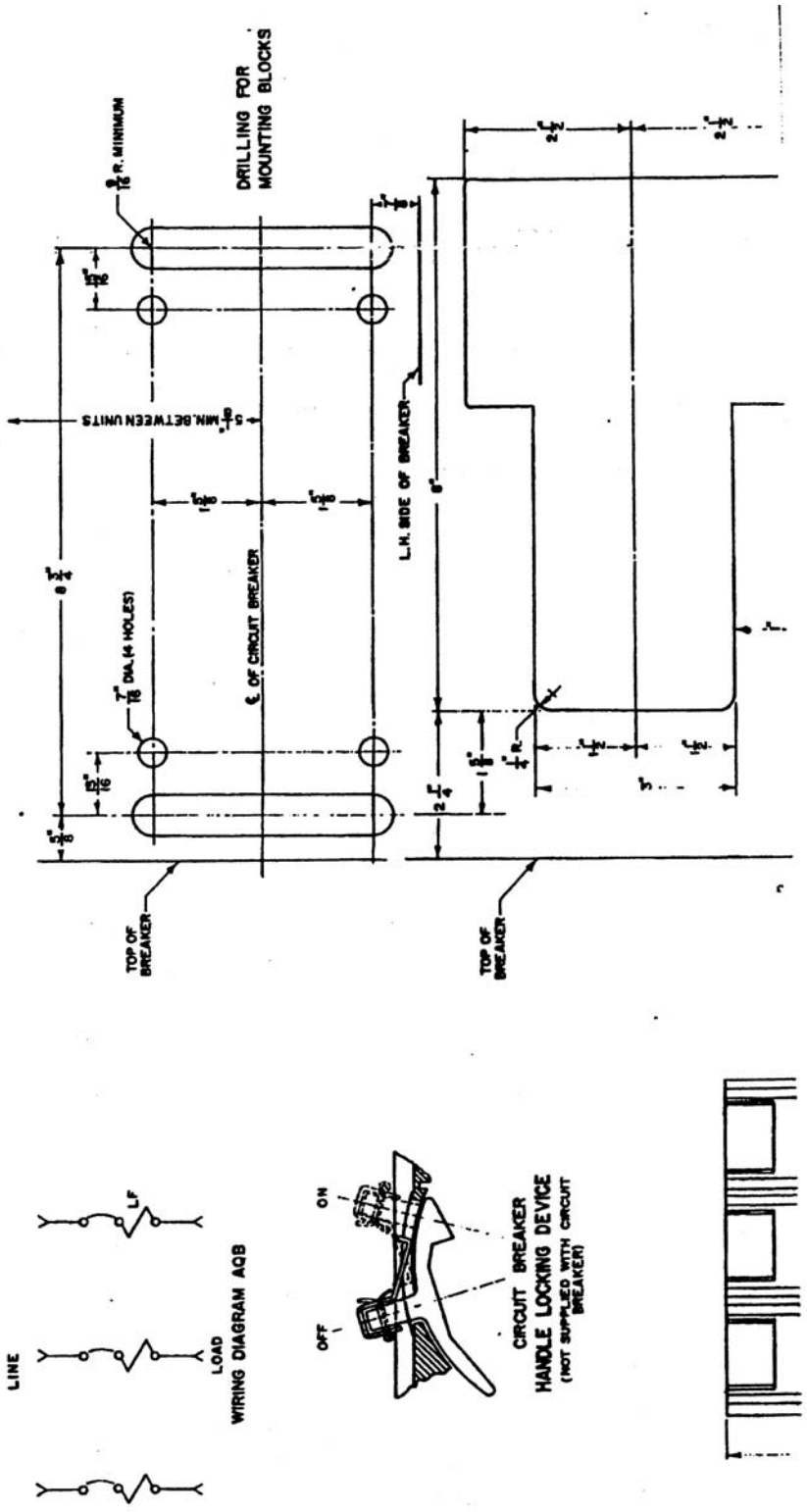


FIGURE 1. Circuit breaker type AQB-LF100, outline and mounting dimensions.

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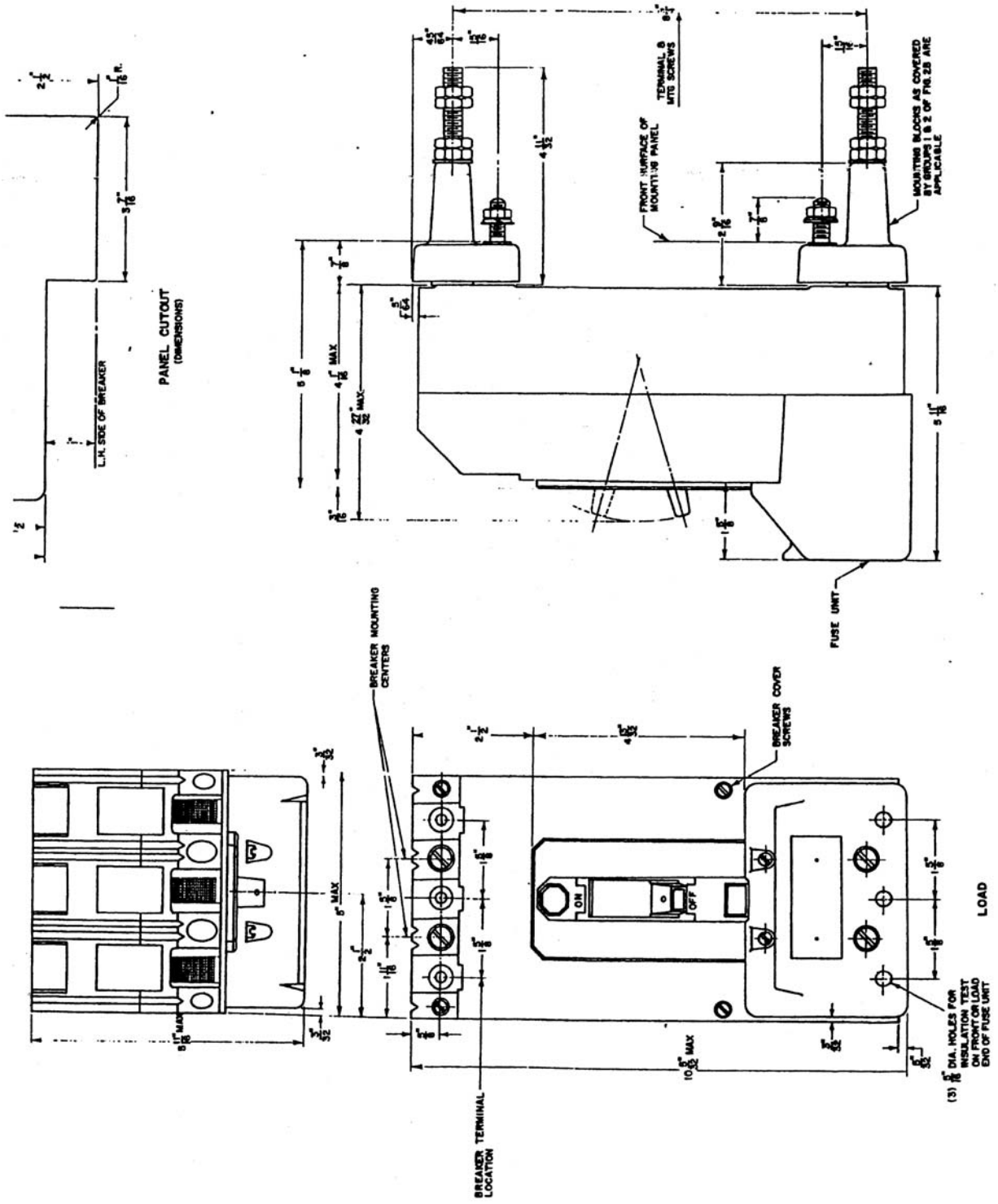
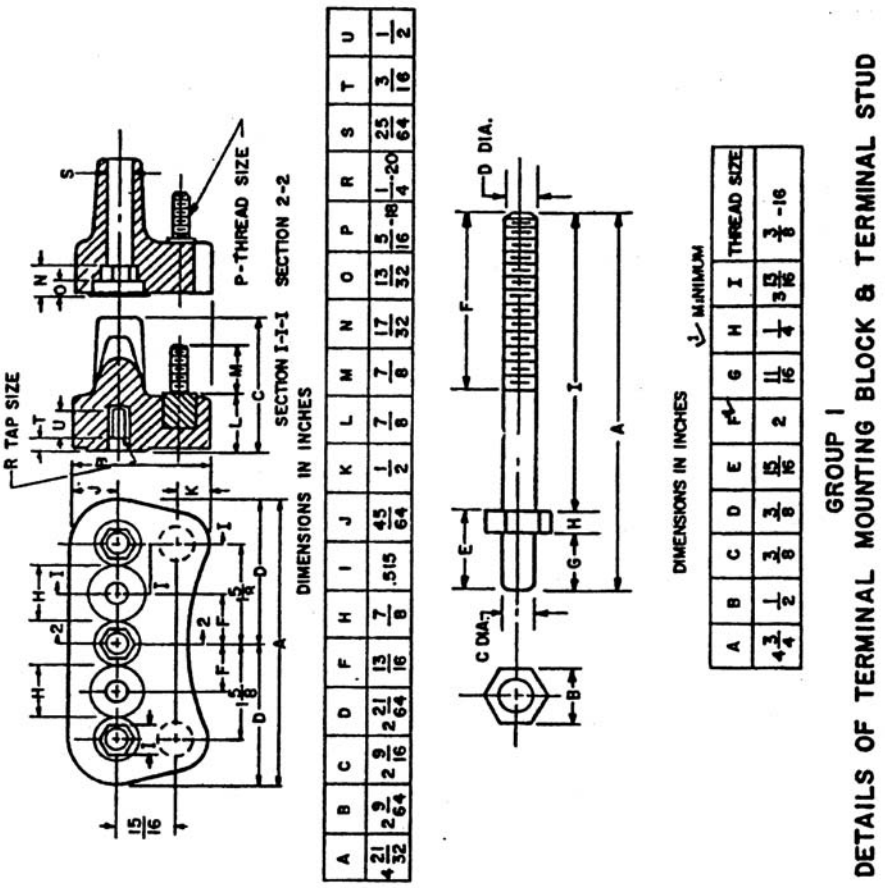


FIGURE 1. Circuit breaker type AQB-LF100, outline and mounting dimensions - Continued.

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A	B	C	D	F	H	I	J	K	L	M	N	O	P	R	S	T	U
4 21/32	2 9/64	2 9/16	2 21/64	13/16	7/8	.515	45/64	1/2	7/8	7/8	17/32	13/32	5/16-18	1/4-20	25/64	3/16	1/2

A	B	C	D	E	F	G	H	I	THREAD SIZE
4 3/4	1/2	3/8	3/8	15/16	2	11/16	1/4	3 13/16	3/8-16

FIGURE 2. Mounting blocks for circuit breaker type AQB-LF100 outline and mounting dimensions.

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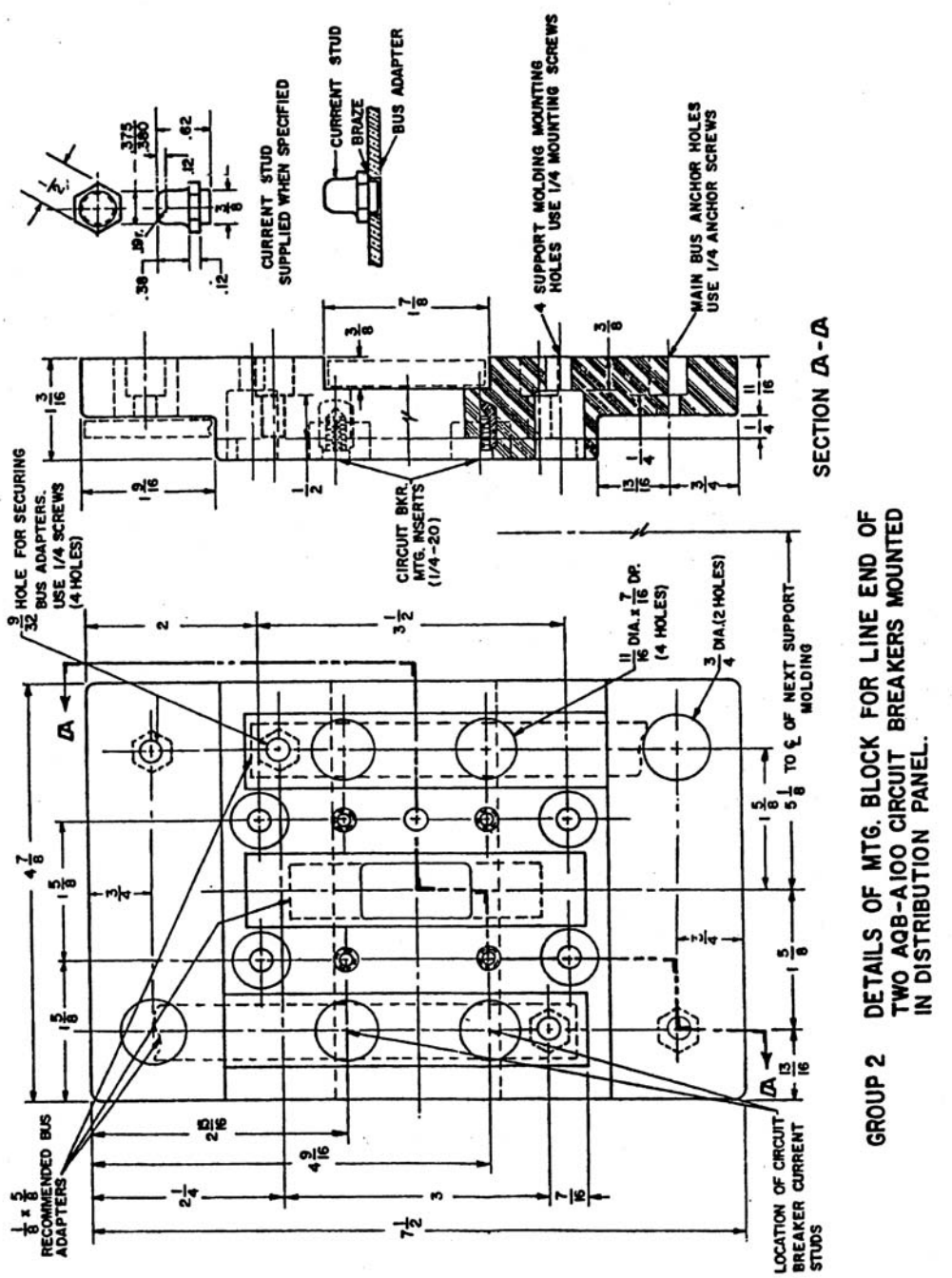


FIGURE 2. Mounting blocks for circuit breaker type AQB-LF100 outline and mounting dimensions - Continued.

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy – SH
(Project 5925-N185-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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